

EAN-File Recording

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 **CAUTION:** Alerts to a condition or improper procedure that can result in equipment or component damage if not avoided.

 **IMPORTANT:** Identifies information that is critical to successful setup, configuration, or operation of the system.

 Provides supplementary information that aids in the use or understanding of the system or subject matter but is not critical to operation.



1 Overview

Sightline hardware can be used to record compressed video and high-resolution snapshots. These can be used for archiving, additional image analytics by Geoinformation System (GIS) software, or other applications in a post-operating environment.

Transferring files from the microSD card to the PC can be done by removing the card from the Sightline hardware and inserting it into the PC. Files can also be remotely accessed using FTP.

This document describes configuring the Sightline hardware to record video or snapshots to either the on-board microSD card, a USB drive, an external FTP drive, or to a mapped network drive. It also covers additional features related to onboard storage.

1.1 File Recording Features

- Snapshots and video can record additional metadata such as latitude and longitude, aircraft position and other related fields in their MISB defined Key-Length-Value (KLV) formats.
- Image quality and compression rates are adjustable.
- Snapshots can be recorded to microSD card, USB drive, FTP server, or mapped network drive.
- Video can be recorded to microSD card, USB drive, or mapped network drive.
- Snapshots can be either JPEG, PNG, or SLRAW.
 - The original capture image or the final display image can be recorded.
 - JPEGs and PNGs are recorded with embedded metadata preserving vital geolocation information with each image.
- Additional Metadata can be recorded to KML (Google Earth), NITF, or KLV files.
 - KML (Keyhole Markup Language) is an XML-based notation for capturing geographic information and allows for accurate importing of still images to 2D and 3D maps within software packages such as Google Earth.
 - NITF (National Imagery Transmission Format) is a common standard file format used within the U.S. DoD and domestic Police and Fire digital imaging community.
 - KLV is a binary block of data that includes everything from vehicle data to target information. This block of data is the same data that is found in the MPEG2-TS stream.
- Video files are recorded as .ts files (transport stream) and contain H.264/H.265 + KLV encapsulated in MPEG2-TS streams.
 - HTTP Live Streaming (HLS) is recorded as .m3u8.

1.2 Additional Support Documentation

The [Panel Plus](#) User Guide is accessible from the Help menu within the application.

The Interface Command and Control (IDD) describes the native communications protocol.

Additional Engineering Application Notes (EANs) and ICDs are available on the [Product Support Documentation](#) page in the [Sightline Knowledge Base](#).



1.3 Sightline Software Requirements

❗ IMPORTANT: The Panel Plus software version should match the firmware version running on the OEM. Firmware and Panel Plus software versions are available on the [Software Releases](#) page.

Table 1: Sightline OEM Software Requirements

OEMs	Initial Software Release	Provisions
4110-OEM	3.11.x	--
1710-OEM	3.9.x	--
4100-OEM	3.7.x	Three channel processing requires 3.9.x and above.
1750-OEM	3.6.x	Dual channel processing and Snap All Cams requires 3.7.x and above.
4000-OEM	3.00.xx	AI-Assist and AI-Detect are not supported. Blending requires 3.3.x and above.
Legacy OEMs are supported up to 3.5.x		
1500-OEM	2.20.xx	--
3000-OEM	2.22.xx	Rev B6 requires updated firmware to accommodate the change in memory.

1.4 Software Feature Version Requirements

Software feature availability is determined by software version. Refer to the table below for the minimum software version that supports each feature described in this document. For questions or upgrades, contact [Sales](#).

Table 2: Software Feature Version Requirements

Recording / Snapshot Features	Initial Software Release
Snapshot Recording (JPEG)	2.22.xx
Snapshot Recording (PNG)	2.24.xx
Full pixel depth lossless snapshot	2.24.xx
Snapshot Recording (SLRAW)	3.00.xx
Video Recording (H.264)	2.22.xx
HTTP Live Streaming (HLS)	2.24.xx
Video Recording (H.265)	3.00.xx
Recording to USB (4000 and 17xx)	3.2.x
Burst Mode Snapshots	3.2.x
Detection Snapshots	3.2.x
KLV (optional)	2.22.xx

1.5 Image Compression Standards

Sightline has implemented both JPEG and PNG standards for capturing video frames, as well as a custom SLRAW format. Snapshots of unprocessed frames from a connected camera and processed frames (display) are supported.

Table 3: Support Digital Compression Standards

Snapshots and Video	Supported Standard
JPEG - Joint Photographic Experts Group	ISO/IEC 10918
PNG - Portable Network Graphics	ISO/IEC 15948
SLRAW - Sightline Raw High Bit Depth	N/A
Video files - .ts files (MPEG transport stream)	ISO/IEC 13818-1
KLV files - KLV metadata	MISB ST 0107



2 Hardware Setup

IMPORTANT: The microSD card or USB drive must be formatted before use. See [Appendix A](#) before recording files to the card. Power off the unit before inserting or removing the microSD card or USB drive.

MicroSD card slot locations and USB connectors are shown in the dimensional drawings for Sightline OEMs available from the [Dimensional Drawings and 3D Models](#) page in the [Knowledge Base](#).

3 File Naming

IMPORTANT: Whenever files are being recorded, it is important to be vigilant during the process to prevent accidental erasure of critical data.

For both snapshots and video recording the software can create new files with a unique suffix based on the syntax of the file, making it easier to identify files or reduce the risk of over-writing existing files.

File name length: The filename length for video and snapshots is limited to 74 characters or less.

Auto numbering: File names are automatically appended with four digits to provide unique file naming. If the base file name ends in a number (0 to 9), auto-numbering will be disabled.

CAUTION: When video is being recorded, if the user given filename ends with a numeral, it will be overwritten when the recording size is reached (1 GB is the default unless defined by the user).

IMPORTANT: If using [Snap All Cams](#) feature it is important that the base file name does not end in a number (0 to 9) or the filenames for the cameras may have the same name.

Camera numbering: (17xx-OEM, 3000-OEM, 4000-OEM, and 41xx-OEM) - The camera number is automatically inserted into the file name unless the user specifies a file name ending in 0 to 9.

Example:

File name: *Landing_0_0000.jpg*

- Camera number = 0
- The first number = 0000
- The next snapshot = *Landing_0_0001.jpg*.
- A snapshot on camera 2 = *Landing_2_0000.jpg*.

When the source is configured for both capture and display, filenames include either **_cap** or **_disp** between the camera number and the snapshot number. For example: *Landing_0_cap_0000.jpg* for a capture snapshot, and *Landing_0_disp_0000.jpg* for a display snapshot.

Table 4: File Naming Conventions

If using Filename:	File name 1500:	File name 17xx/3000/4000/41xx:	Notes
Hello	Hello_0000.jpg	Hello_0_0000.jpg	Count starts at 0
Hello	Hello_0001.jpg	Hello_0_0001.jpg	Count increments
World	World_0000.jpg	World_0_0000.jpg	Count restarts when Filename changes
Hello42	Hello42.jpg	Hello42.jpg	Count is not appended for Filename ending in a numeric character.



Detection Snapshots:

The naming convention for detection snapshots is similar to single frame snapshots except for the following changes:

- The track ID corresponding to the detection will be appended to the filename.
- The base filename cannot end in numbers 0 to 9.
- All snapshots taken on the same frame will have the same snapshot number.
- To replace the snapshot number with the 8-byte UTC timestamp, select the *Use Timestamp in Filename* checkbox.

Example:

File name: *Snapshot_0_0001_0010.jpg*

- Camera number = 0
- Snapshot number = 0001
- Track index of the detection = 0010
- A snapshot for track index 11 on the same frame = *Snapshot_0_0001_0011.jpg*

4 Configure Snapshot Source, Destination, and Formats

Video and snapshots can be recorded to different destinations (FTP, microSD, USB, etc.). In Panel Plus go to the *Record* tab » *Advanced Settings* to access the *Configure Snapshot Settings* dialog.

IMPORTANT: Click *Send* when making any setting changes in the dialog window.

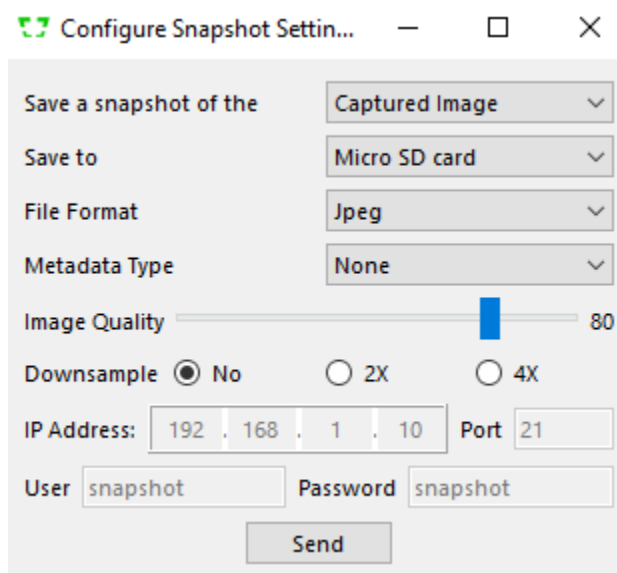


Figure 1: Configure Snapshot Settings



Save Snapshot (Capture/Display)	<i>Captured Image:</i> The original unprocessed image will be saved. <i>Display Image:</i> Saves a processed image. This includes stabilization, digital zoom, overlay graphics, false coloring, and image enhancement. <i>Both:</i> Saves two images: one before Sightline processing and one after. <i>When using Display Image or Both, the source video should be displayed on a network or on an HDMI output.</i> <i>When Both is selected, the Capture Snap File Format and the Display Snap File Format must be specified.</i>
Save To (File Destination)	Determines where the images will be saved to: <i>FTP Server:</i> Files are sent to the FTP server. See FTP Server Settings. <i>Micro SD Card:</i> Files are written to the microSD card. <i>USB Drive:</i> Files are written to the USB drive (17xx-OEM, 4000-OEM, and 41xx-OEM only).
Image Quality	Keep JPEG quality below 90 on the Image Quality sliding scale. Higher settings result in large files that slow down the snapshot rate. Since PNG is a lossless compression the sliding scale only controls the file size and not the image quality. Higher settings result in a smaller sized file. Settings below 80 are not recommended and will result in a minimally smaller file with increased processing time and a lower snapshot rate.
Metadata Type	<i>None:</i> No Metadata file is saved. <i>Single KML File:</i> One single KML file is saved when taking snapshots. During multiple snapshots, the metadata for all the snapshots will be contained in a single file. <i>Multiple KML Files:</i> Every snapshot creates a new KML file. <i>NITF File:</i> Snapshot with metadata is stored in an NITF file (National Imagery Transmission Format). Compliant with STANAG 4545 as of 3.10.x. <i>KLV (new for 3.2.xx):</i> Every snapshot creates a KLV file. This file contains a binary block of KLV data as defined in EAN-KLV-Metadada. SLA-Decode example can be used to decode the data from this file. <i>When using KLV metadata in combination with detection snapshots, detection snapshots will be limited to targets included in the KLV metadata.</i>

4.1 File Format Options

JPG and PNG file types can also contain metadata. An EXIF header is added to JPEG files. PNG files will have a custom EXIF-like header added on.

The JPEG file format is not supported for 14-bit or 16-bit digital cameras. Use the PNG or SLRAW file format when recording the captured image. Set snapshot settings to Displayed Image to use other file formats.

Table 5: File Format Options

File Format	Description	Compression
JPG	Joint Photographic Experts Group (JPEG)	Lossy compression
PNG	Portable Network Graphics (see PNG)	Lossless compression
JPG Big (new in 2.24.xx)	Same as JPEG/JPG. Allow a larger snapshot to be taken of capture (source) image (see Big Snapshot).	Lossy compression
SLRAW (.slraw)	Sightline proprietary format that does not encode the image buffer (see SLRAW).	Lossless compression
TIFF	Uncompressed images with a standard tiff wrapper for 16-bit grayscale capture snapshots only. (See TIFF)	Uncompressed



4.1.1 EXIF Headers

The following fields are typically output with a JPEG in the EXIF header. For georeferenced tagging, KLV must be enabled and the values must be supplied using the [SetMetadataValues \(0x13\)](#) or [MetadataStaticValues \(0x14\)](#) commands, or by enabling NMEA 0183 parsing on the serial port using the [SetPortConfiguration \(0x3E\)](#) command. See the NMEA section in [EAN-Ethernet-and-Serial-Communication](#).

See [Appendix E](#) for a table of EXIF headers.

To see EXIF additions, right-click on the JPEG file and choose Properties.

Saved image EXIF Tag provisions are included but are not implemented to meet the requirements of any specific EXIF standards version.

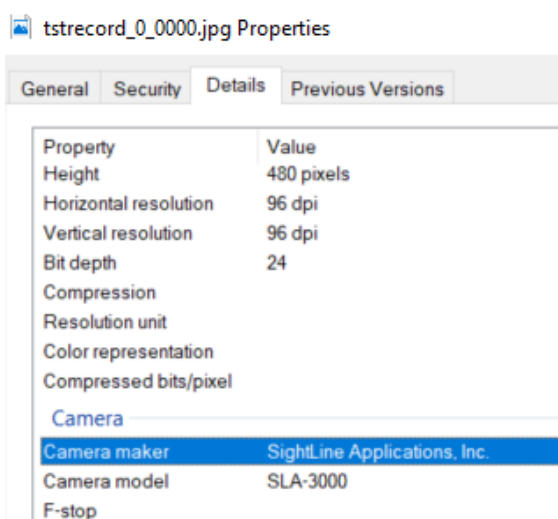


Figure 2: Example EXIF Header in JPEG

4.2 FTP Server Settings

When saving to FTP server set the following parameters:

- **IP Address:** IP address of the remote server machine.
- **Port:** Network port the FTP server listens for connections.
- **User and Password:** Username and password required for login.

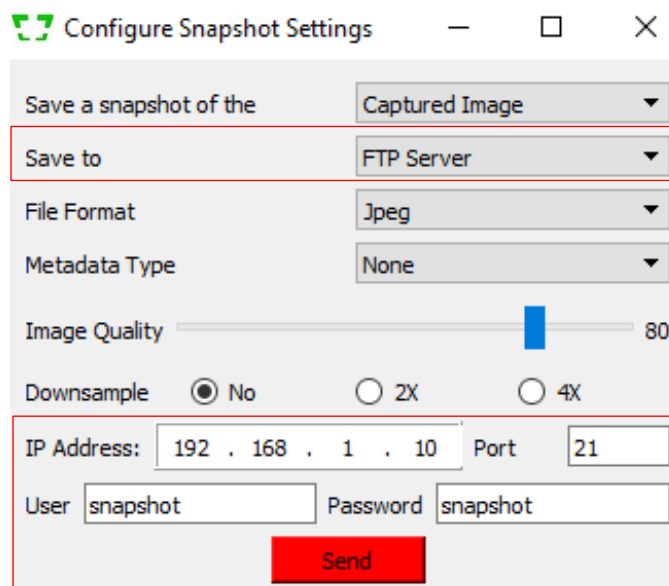


Figure 3: Recording to FTP Server Settings



5 Single Frame Snapshot

In Panel Plus select the *Record* tab » *Single Frame (Snapshots)* tab.

Figure 4: Single Frame (Snapshots) Tab

<i>Number of snapshots to save:</i>	Represents the number of frame snapshots to take (default set to 1).
<i>Continuous:</i>	Checking this box will enable continuous snapshots. One snapshot will be taken for each frame based on the <i>Frame Step Between snaps</i> value.
<i>Frame steps between snaps:</i>	Step between frames, e.g., 2 = snap every other frame where default is set to 1.
<i>Snap All Cams:</i>	Allows for a single snapshot to be taken by two camera inputs at the same time. See Snap All Cams for more information on using this feature.
<i>Max files per folder:</i>	Allows users to control the number of snapshots stored in each folder. This is a maximum number of files per folder. The folder may auto increment sooner under certain conditions, e.g., snapshots taken too fast.
<i>Auto Folder:</i>	Without breaking the data up into additional folders, system performance will start to slow down at ~2000 files. The maximum file capability is limited to ~22,000 files. The <i>Auto Folder</i> option enables more files to be taken with minimal impact to performance. Sightline has tested to up to 500,000 snapshots. During auto-numbering, if the file name is changed the system will automatically scan the microSD card to find the highest number. It will continue autonumbering from that number forward. This helps avoid overwriting files that were previously written. Changing the Auto Folder state will also cause the system to scan the microSD card again.
<i>Burst Mode:</i>	Allows for a burst of snapshots to be taken without dropped frames or frame skipping. The maximum burst depends on the speed of the recording device, size of the image, and other factors. A maximum of 120 images up to 1920x1080 can be burst on slower microSD cards. This feature is not supported with Snap All Cams or with images greater than 1920x1080 pixels (17xx-OEM, 4000-OEM and 41xx-OEM only).
<i>Base file name of saved files:</i>	See File Naming .
<i>Start/Stop:</i>	Sends the DoDetectSnapShot (0xAB) command that prompts the hardware to start/stop taking snapshots using the parameters provided.

Files will be recorded on the microSD card if there is available memory. When the card is full recording will stop until more memory is free.



6 Detection Snapshots

In Panel Plus go to the *Record* tab » *Detection Snapshots* tab.

Figure 5: Detection Snapshots Tab.

<i>Snapshot Size Type:</i>	Select how the size of the snapshot file will be calculated: • <i>Fixed Size:</i> Square of <i>Snapshot Size</i> centered on the detection. • <i>Scaled:</i> The actual size of the detection is scaled up or down to <i>Snapshot Size</i>. • <i>Auto:</i> The size of the snapshot is based on the detection size and the <i>Snapshot Size</i> is ignored.
<i>Snapshot Size:</i>	Controls the size of the output snapshot along with <i>Snapshot Size Type</i> . The minimum size is 32 and the maximum is 512. Sizes must be a multiple of 16.
<i>Max Snapshots Per Frame:</i>	Adjust how many snapshots to save per frame. Depending on the speed of the device that snapshots are being written to, setting a number too high can result in snapshots being dropped. Sightline recommends using a small number for this setting since there is logic in the detection snapshot recording that cycles through all the detections.
<i>Min Frames Between Snapshots:</i>	Adjust the minimum spacing in frames between snapshots of the same detection or track. For example, if a snapshot of a target is needed every 1 second, and the video is 30 Hz, set this number to 30.
<i>Use Timestamp In Filename:</i>	See File Naming .
<i>Base file name of saved files:</i>	See File Naming .
<i>Start/Stop:</i>	Sends the DoDetectSnapShot (0xAB) command that prompts the hardware to start/stop taking snapshots using the parameters provided.

7 Big Snapshot

Big snapshot is a feature used for higher resolution cameras to run normal video processing on a lower resolution stream but take a high-resolution snapshot. Sightline OEMs have different implementations and capabilities due to hardware and encoder differences.

**Table 6: Maximum Resolution for Big Snapshot**

OEM	Snapshot Resolution	Streaming Video Resolution
1500-OEM	1080	480 or 720
3000-OEM	20 MP	1080
17xx-OEM/4000-OEM/41xx-OEM	16 MP	4K

During a maximum resolution snapshot on the 17xx-OEM, 3000-OEM, 4000-OEM and 41xx-OEM, the video display is stopped and then returns when the snapshot is complete.

7.1 Camera Configuration Notes

Maximum image size for 3000-OEM big snapshot:

- Bayer and 8-bit gray scale: 20MP image
- YUV: 13MP image.

Maximum image size for 17xx-OEM big snapshot:

- 10-bit Bayer: 12MP image
- 8-bit Bayer and 8-bit gray scale: 16MP image
- YUV: 8MP image

Maximum image size for 4000-OEM and 41xx-OEM big snapshot:

- 10-bit Bayer: 12MP image
- 8-bit Bayer and 8-bit gray scale: 16MP image
- YUV: 8MP image

7.2 1500-OEM Big Snapshots

The 1500-OEM can capture up to 1080 resolution snapshots while streaming SD or 720p video without interrupting the video stream. This is accomplished using the *Region of Interest* setting in the *Acquisition Settings* dialog window.

This method is not supported in the 3000-OEM, 4000-OEM, and 41xx-OEM.

7.3 1500-OEM Camera Configuration

1. Connect to the board using the Panel Plus application.
2. From the main menu in Panel Plus » *Configure* » *Acquisition Settings*.
3. Set up the digital camera for full resolution capture. In the example shown in **Figure 6**, the *Camera Index* is set to *Digital*, and the *Camera Type* is set to *Sony FCB-EH 1080P* (1080x1920 resolution).
4. Set the *Region of Interest* to the resolution that will run in the normal video stream. In this case, *640x480* video.
5. Click the *Apply* button.
6. Click the *Record* tab and then click *Advanced Settings* to open the *Configure Snapshot Settings* dialog window.

Figure 6: Big Snapshot Acquisition Settings - 1500-OEM



7. Set the fields as shown in **Figure 7**.
8. Click the *Send* button.
9. From the *Record* tab click the *Start* button under *Single Frame (Snapshots)* tab. A 1920x1080 resolution snapshot will be recorded.

Figure 7: Configure Snapshot Settings - 1500-OEM

7.4 3000-OEM Big Snapshot

The 3000-OEM version of big snapshot allows the user to interrupt normal video processing and streaming to capture an up to 20MP resolution snapshot. When the camera is configured to use the big snapshot feature, during a snapshot, acquisition is stopped and reset using the big snapshot settings. When the big snapshot is complete, acquisition is again stopped and reset to use the original acquisition settings. During big snapshot video display is stopped and then returns when big snapshot is complete.

7.5 3000-OEM Big Snapshot Camera Configuration

Generic Digital is the only camera type that is compatible with the big snapshot feature.

Configure the camera to the resolution that will be used for taking large snapshots. The maximum resolution for large snapshots is 20MP. For an 8-bit gray scale or Bayer camera, this corresponds to 20MP. For a YUV or 16-bit gray scale camera, the maximum pixel resolution is reduced by the bytes per pixel.

1. Connect to the board using the Panel Plus application.
2. From the main menu in Panel Plus » *Configure* » *Acquisition Settings*. The *Apply* button will turn red indicating input field changes have been detected.
3. Set the camera resolution to a lower resolution that will be used in the display. In the example shown in **Figure 8**, the *Camera Index* is set to *Cam 0*, and the resolution is set up to capture at 1080x1920.



- ❗ IMPORTANT:** The *Camera Type* must be *Generic Digital* for the big snapshot feature to work correctly.

Figure 8: Big Snapshot Configuration Example - 3000-OEM

4. In Panel Plus go to the *Record* tab » *Advanced Settings* to open the *Configure Snapshot Settings* dialog window.
5. Set the fields as shown in **Figure 9**.
6. Click the *Send* button.
7. From the *Record* tab click the *Start* button under *Single Frame (Snapshots)* tab. A 20MP snapshot will be recorded on the microSD card.

Figure 9: Configure Snapshot Settings - 3000-OEM

7.6 17xx-OEM, 4000-OEM, 41xx-OEM Big Snapshot

The 17xx-OEM, 4000-OEM, and 41xx-OEM version of the big snapshot feature allows the user to interrupt normal video processing and streaming to capture a resolution snapshot up to 16MP. During a snapshot only the processing of the snapshot occurs until the snapshot is saved, and then normal processing continues.



7.7 17xx-OEM, 4000-OEM, 41xx-OEM Big Snapshot Camera Configuration

Generic Digital, USB Webcam, and USB3 Vision cameras work with the big snapshot feature.

Configure the camera to the resolution that will be used for taking large snapshots. The maximum resolution for large snapshots is 4000x3000.

1. Connect to the board using the Panel Plus application.
2. From the main menu in Panel Plus » *Configure* » *Acquisition Settings*. The *Apply* button will turn red indicating input field changes have been detected.
3. Set the camera resolution to the *Height* and *Width* of the big snapshot as shown in **Figure 10**.

Use the ROI parameters for display size. In this example, the ROI is set to display a 1280x720 image. When a big snapshot occurs the *Width* and *Height* of the snapshot will be 4000x3000.

Acquisition Settings

Camera Index: Cam 0

Camera Type: Generic Digital

Frame Step: 0

Board Firmware Ver: 3.9.2.2 FPGA:00:00

Generic Digital Settings

Auto Fill:

Height: 3000 Width: 4000 Resulting Flag Bits: 0x3

Vertical Front Porch: 0 Horizontal Front Porch: 0 Bit Depth: 10

Input: ☐ Gray Scale ☐ YUV color ☐ G8 16bit in ☒ Bayer ☐ Laser ☐ Interlaced ☐ Byte Swap

☐ Invert V-Sync Polarity ☐ Invert H-Sync Polarity ☐ UB0 ☐ 2xbins Sync/Crop: None

Camera Init Code: None

Options: mipi=imx412

Big: Height: 0 Width: 0 Vertical Blanking: 0 Horizontal Blanking: 0

Region of Interest (Advanced Setting)

First Valid Row: 1140 Valid Height: 720 Centered ROI

First Valid Column: 1360 Valid Width: 1280 1280x720

Figure 10: Big Snapshot Configuration Example - 4000-OEM/41xx-OEM



1. In Panel Plus go to the *Record* tab » *Advanced Settings* to open the *Configure Snapshot Settings* dialog window.
2. Set the fields as shown in **Figure 11**.
3. Click the *Send* button.
4. From the *Record* tab click the *Start* button under *Single Frame (Snapshots)* tab. A big snapshot will be taken using the *Width* and *Height* settings.

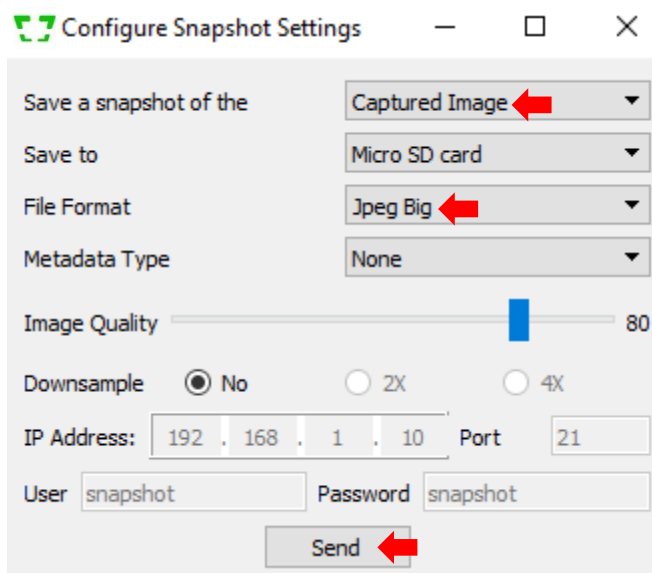


Figure 11: Configure Snapshot Settings - 4000-OEM/41xx-OEM

7.8 16-Bit Snapshots

When working with an infrared camera as outlined in [EAN-Infrared-Temperature](#), recording snapshots of the raw 16-bit camera output can be useful. Sightline supports 16-bit recording in either PNG, SLRAW, or TIFF format, which are described in the following sections.

 **3000-OEM and 1500-OEM:** When performing a NUC calibration, and the camera is configured with a bit depth of 16, the pixel values will be divided by 2. See [EAN-NUC-and-DPR](#) for more information.

7.8.1 PNG

Key features of the PNG format are described below:

- Lossless compression
- 16-bit grayscale PNG files can be displayed in most image viewer software:
 - Viewer software normally only displays the top 8-bits resulting in images that appear mostly black with faint imagery.
 - Software such as [Image-J](#) can be used to view the complete 16-bit image.
- Also supports RGB data so it can be used for capturing 16-bit grayscale or display 8-bit RGB snapshots.
- Supports user chunks of binary data (this is where MISB data is stored).
- 16-bit PNG files can be imported into MATLAB via `imread()` function.

7.8.2 SLRAW

The following are key features of the SLRAW format:

- Sightline proprietary format that does not compress the image buffer.
- Faster write speeds compared to PNG.
- No support for user chunks of binary data.
- No support for display snapshots.



The SLRAW format image consists of a 12-byte header followed by uncompressed raw image pixel data.

The header consists of the following bytes (little endian order):

- 4-byte identifier = 0x51acd00d
- 4-byte image type identifier. Supported options include:
 - 5 = 8-bit grayscale - 8-bit single byte value per pixel.
 - 6 = 16-bit grayscale - 16-bit little endian order 2-byte value per pixel.
 - 21 = RGGB10 Bayer format image stores 4 pixels in 5-bytes. The first 4-bytes are the high 8-bits. of the 4 pixels and the fifth byte packs the bottom 2-bits from the 4 pixels.
 - Even rows repeat RGRGX where X is the packed low 2-bits from RGRG.
 - Odd rows repeat GBGBX where X is the packed low 2-bits from GBGB.
- 2-byte image width
- 2-byte image height

 *Raw pixels described above immediately follow the header.*

Software such as [ImageJ](#) can be used to view the complete 16-bit image using the following steps:

1. Select *Menu File » Import » RAW*.
2. Manually enter the image width and height in the appropriate fields.
3. Enter 12 bytes in the *Offset to first image* field.
4. Check the *Little-endian byte order* box.

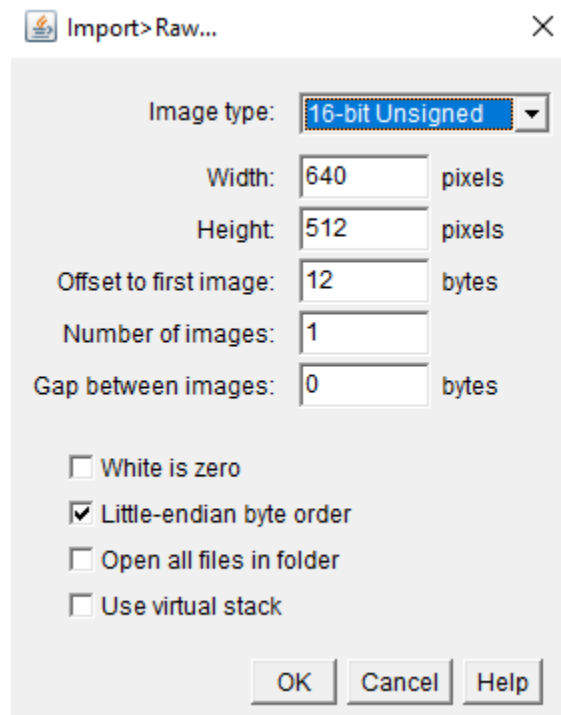


Figure 12: ImageJ Software Import Example



7.8.3 TIFF

The following are key features of the TIFF format:

- Uncompressed images
- 16-bit grayscale TIFF files can be displayed in most image viewer software:
 - Does not currently support color images.
 - Does not currently support binary data, e.g., GeoTIFF.

7.8.4 Sample Application - Viewing 16-Bit Images

SLA-PNGViewer is a sample application for viewing the full 16-bit data from either PNG or SLRAW files. The sample application is built upon the sample code SLADecode, which can be downloaded from the Sightline [Command and Control](#) page. Highlights include:

- Easily ported to Linux. Includes *libpng* and *zlib* source (open source).
- Decodes and auto-scales PNG 16-bit and SLRAW 16-bit data for display. Displays temperature data.
- Extracts MISB KLV metadata user chunk from PNG 16-bit and decodes sensor latitude and longitude for display as an example.
- Includes an example 16-bit PNG takes from FLIR Tau 2 TLinear with MISB KLV included.

 For more information contact [Support](#) for a demonstration video of the decode process.

7.8.5 Snapshot Rates

Capturing high-rate high bit-depth snapshots is often time important when working with infrared cameras. See [Snapshot Performance](#).

8 Snap All Cams

Since the 3000-OEM, 17xx-OEM, 4000-OEM, and 41xx-OEM support two simultaneous network streams, the *Snap All Cams* feature can be used to take snapshots from both camera inputs simultaneously. When using this feature, care must be taken in configuring each camera separately by specifying the command camera when setting configurations.

 **IMPORTANT:** *Snap All Cams* for the 1750-OEM requires software version 3.7.x and above.

Snap All Cams using Panel Plus:

1. Set up dual network streams. See the [Recording Multiple Streams](#) section.
2. In Panel Plus set the command camera to 0.
3. From the *Record* tab click the *Single Frame (Snapshots)* tab.
4. Check *Snap All Cams*.
 - ☐ Continuous
 - ☒ Snap All Cams
5. Specify a filename for command camera 0.



The Snap All Cams feature works with both FTP and SD Recording.

6. Set the command camera to 2.
7. Click *Advanced Settings* and configure the settings.
8. Set command camera back to 0.
9. Click *Start*. The following two snapshots are saved to the specified destination:
 - `<filename>_0_<fileNumber>` (0 represents the camera index)
 - `<filename>_2_<fileNumber>` (2 represents the camera index)

9 Video Recording

IMPORTANT: Version 3.07.xx and above:

- RTP MPEG2-TS (H.264 or H.265) streams can be recorded much similarly to MPEG2-TS streams. The RTP header is removed, and the MPEG2-TS data is recorded with a `.ts` extension containing KLV and Video data.
- Previous versions treated RTP MPEG2-TS as an RTP stream and would only record an un-encapsulated H.264 stream. See below for more details.

IMPORTANT: Version 3.01.xx and above:

- When recording video, use MPEG2-TS for the best results.
- If recording an RTP or RTP-MPEG2TS, only the un-encapsulated H264 or H265 video is recorded to the microSD card with a `.h264` or `.hevc` extension. This data can be played back in VLC media player, Windows Media Player, or with FFmpeg. See the [Troubleshooting](#) section for more information on how to work with these un-encapsulated files.
- Video recording of RTP MJPEG is not currently supported.
- Recording videos with a very low frame count or size may produce unplayable video. Lowering the I-Frame Interval in the *Capture* tab may allow the recording of smaller videos.

IMPORTANT: Version 3.00.xx and under:

- The video recording is limited to MPEG2-TS video. Only the same encoding format that is being transmitted over Ethernet can be recorded, e.g., streaming H.264 video while recording H.264 video.
- Recording RTP transport stream video is not supported. RTP streaming is only used for real-time video over IP networks. Record and playback are not supported on most video players. RTP streams are typically converted and recorded by a ground station utilizing a video management system.

1. To start video recording, click the *Compress* tab in Panel Plus and set the encoding format.

CODEC / TRANSPORT	
RTP:	☐ MJPEG
	☐ RTP H.264
MPEG2-TS:	☒ H.264
	☐ MPEG4



- Click the *Record* tab.
- Choose where to save the recording. Click the *Video* tab under *Onboard Recording* or under *Panel+ (PC)Recording*.
- Click the *Start* button to start recording...

Records to microSD card →

17xx-OEM/4000-OEM/41xx-OEM →

When recording video with the 17xx-OEM, 4000-OEM, and 41xx-OEM ensure the network specified in the *Recording* tab is configured correctly for the camera(s) in the camera matrix in the *Display* tab.

Recording Status →

Records to host PC →

Figure 13: Record Tab Video Recording

- When the recording is started and stopped file names that do not end with a number are automatically appended with a suffix with consecutive numbers. Most video files are recorded with the .ts suffix (transport stream) and can be played back with Windows Media Player, VLC, or others. However, if recording RTP the suffix will not be .ts. See the [Troubleshooting](#) section for more information.
- The default setting for video file recording is ~1GB. This setting can be adjusted as needed.
- When starting a new file every x-number of frames, the actual number of frames in the recorded file may be larger than specified up to the next i-frame.



9.1 Recording Multiple Streams

The 17xx-OEM, 4000-OEM, and 41xx-OEM can record two videos simultaneously. These correspond to the video streamed over *Net 0* and *Net 1* and must be set up in the *Compress* tab.

These are logical network channels used to generate an outbound stream. The same rules apply for defining a destination address for either a single host or multiple hosts. If sending both streams to the same host, set a unique port address for each stream shown in **Figure 14**.

Set up each recording separately. One to Record Net 0 and one to record Net 1.

Figure 14 shows two network configuration windows. The top window is titled "Network 0 - Defined" and the bottom window is titled "Network 1 - Defined". Both windows have a "Streaming" section with a "To IP Address" field set to "192.168.0.31" and a "Port" field. In the "Network 0" window, the "Port" is "15004". In the "Network 1" window, the "Port" is "15008". A red arrow points from a text box "Ports should be unique/different if destination IP is the same." to the "Port" fields of both networks.

Figure 14: Output Separate Streams - 3000-OEM, 4000-OEM, and 41xx-OEM

They should correspond with the Network definitions above.

9.1.1 Recording Multiple Streams from Factory Defaults

See one of the corresponding OEM startup guides for connection and video streaming instructions:

- [EAN-Startup-Guide-1710-OEM](#)
- [EAN-Startup-Guide-4000-OEM](#)
- [EAN-Startup-Guide-4100-OEM](#)
- [EAN-Startup-Guide-4110-OEM](#)

1. From the main menu in Panel Plus » *Configure* » *Acquisition Settings* to configure capture parameters for input(s) source.

For information about Acquisition fields in Panel Plus see [EAN-Digital-Video-Configuration](#).

2. In the *Compress* tab, select a supported *Codec/Transport* standard.

Figure 15 shows the "CODEC / TRANSPORT" selection form. It has two main sections: "RTP" and "MPEG2-TS". Under "RTP", there are four radio buttons: "MJPEG", "RTP H.264", "RTP H.265", and "RTP MPEG2-TS H.264". Under "MPEG2-TS", there are two radio buttons: "H.264" (which is selected) and "H.265".



- In the *Streaming* section choose a streaming destination. In this example the *Use my IP - Unicast* function is used for *Network 0* and *Network 1*.

Network 0

Streaming

To IP Address: 192.168.1.21 Port: 15004

☒ Use My IP - Unicast ☐ Use Multicast ☐ Broadcast

☐ Start/Stop All Nets

Network 1

Streaming

To IP Address: 192.168.1.21 Port: 15006

☒ Use My IP - Unicast ☐ Use Multicast ☐ Broadcast

☐ Start/Stop All Nets

- Navigate to the *Display* tab and enable dual network streams for input(s).

Network Index	Video Source	Network Display		Physical Display		Decode @ P+
		Enable	Resolution	HDMI	Resolution	
0	Cam 1	☒	Out=In	☐	1080p60	☒
1	Cam 0	☒	Out=In	☐	1080p60	☐

Cam0 can also be used as the Network 1 index.

- Navigate to the *Record* tab. From the *Video* tab in the *SD Recording* section select the *Record Net 0* and *Record Net 1* button. Create a unique filename in the *Filename* field for *Network 0* and *Network 1* then click *Start*.

Panel Plus will provide visual feedback when recording in the *Status* dialog.

Video Single Frame (Snapshots) Detection Snapshots

☒ Record Net 0 ☒ Record Net 1 ☐ Record Net 2

Start a new file every 1000 units. ☐ One File Only

Unit Type: ☐ KB ☐ MB ☐ Frames

☒ Default File Size (1 GB)

☒ Resume file numbering from highest on sd card

Status

Net 0 Recording: **ON** (Cam 0) Net 1 Recording: **ON** (Cam 1)

Net 2 Recording: **OFF** (Cam 0)

Net 0 Net 1 Net 2

Filename: Cam0netZero Create a unique file name.

Net 0 Net 1 Net 2

Filename: Cam1netOne Create a unique file name.



9.1.2 Three Stream Recording (41xx-OEM)

Software version 3.9.x adds support for three network streams on the 41xx-OEM shown as shown in Figure 15 and Figure 16 in Panel Plus.

Network Index	Video Source	Network Display		Physical Display		Decode @ P+
		Enable	Resolution	HDMI	Resolution	
0	None	☐	Out=In	☐	1080p60	☐
1	Cam 0	☒	Out=In	☐	1080p60	☐
2	Cam 1	☐	Out=In	☐	1080p60	☒

☐ Separate Network/HDMI resolutions LVDS N/A

Figure 15: Three Network Streams - Panel Plus Display Tab

Onboard Recording

Video Single Frame (Snapshots) Detection Snapshots

☒ Record Net 0 ☒ Record Net 1 ☒ Record Net 2

Start a new file every 1000 units. ☐ One File Only

Unit Type: ☐ KB ☐ MB ☐ Frames

☒ Default File Size (1 GB)

☒ Resume file numbering from highest on sd card

Status

Net 0 Recording: **ON** (Cam 0) Net 1 Recording: **ON** (Cam 1)

Net 2 Recording: **ON** (Cam 0)

Figure 16: Recording Three Network Streams - Panel Plus Record Tab

10 Accessing On-Board Recorded Files

Files recorded to microSD card can be accessed in several ways. Once the hardware is powered off, the microSD card may be removed and connected into the PC using a microSD card reader or similar device. The files can also be accessed remotely using an FTP client or SCP.

Another option is to use an emulator program (Tera Term recommended) to establish an SSH session to the target:

- Login using the default username and password:
 - 1500-OEM and 3000-OEM: *root*
 - 17xx-OEM, 4000-OEM, 41xx-OEM: *slroot*
- Type: `mount`

```
DM-37x#
DM-37x#
DM-37x#
DM-37x# mount
rootfs on / type rootfs (rw)
/dev/root on / type yaffs2 (rw,relatime)
proc on /proc type proc (rw,relatime)
sys on /sys type sysfs (rw,relatime)
tmpfs on /dev/pts type tmpfs (rw,relatime,mode=755)
/dev/mmcblk0p1 on /mnt/mmcblk0p1 type vfat (rw,nosuid,nodev,uid=0,gid=0,dmask=0000,allow_utime=0022,codepage=cp437,iocharset=utf8,quiet,flush,errors=remount-ro)
devpts on /dev/pts type devpts (rw,relatime,gid=5,mode=600)
shm on /dev/shm type tmpfs (rw,relatime)
rwfs on /mnt/rwfs type tmpfs (rw,relatime,size=512k)
rwfs on /tmp type tmpfs (rw,relatime,size=512k)
rwfs on /var type tmpfs (rw,relatime,size=512k)
DM-37x# cd /mnt/mmcblk0p1/
DM-37x# ls
0f02300c015a864b.license 1800500d0155b214.license 1800601c0155b214.license
1701000a0155b214.license
```

Figure 17: Example - Accessing MicroSD Card Files with Tera Term- 1500-OEM



3. For the 1500-OEM type:

```
cd /mnt/mmcblk0p1
```

For the 4000-OEM, 41xx-OEM, and 3000-OEM type:

```
cd /media/mmcblk0p1
```

For the 17xx-OEM type:

```
cd /media/mmcblk1p1
```

4. To list the directory type: `ls`

IMPORTANT: When accessing recorded files, it is important to understand the following system fundamentals:

- The system can stall for several seconds while writing the video file to the microSD card when the stop recording command is issued (*byte 4*).
- The system can stall for several seconds when files are deleted (*byte 5*).
- Do not attempt to access the files while the system is recording.
- Snapshots can be taken at the same time as video is being recorded.
- See the [EAN-Infrared-Temperature](#) document for information on 14-bit images and PNG files.

11 Snapshot Performance

The following tables show the maximum snapshot rates for a microSD card (all platforms) or USB SSD Drive (17xx-OEM, 4000-OEM, and 41xx-OEM). For bursts of 120+ frames on the 17xx-OEM, 4000-OEM, and 41xx-OEM see the [Single Frame Snapshot](#) section.

 Rates are dependent on the microSD card or SSD model, card formatting, image content, and processing load.

Table 7: 1500-OEM MicroSD Card Snapshot Performance

File Format	SD 640x480	720p (1280x720)	1080p (1920x1080)
JPEG 80%	3.5 per second	3 per second	2 per second
PNG (color)	1 per second	1 per second	1 every 3 seconds
PNG (16 bit)	3 per second	N/A*	N/A**
SLRAW (16 bit)	6 per second	N/A*	N/A**
TIFF (16 bit)	6 per second	N/A*	N/A**

Table 8: 3000-OEM MicroSD Card Snapshot Performance

File Format	720p (1280x720)	1080p (1920x1080)	5 MPix resolution
JPEG 80%	3 per second	1 per second	N/A**
PNG (color)	1 every 1.5 seconds	1 every 2 seconds	N/A**
JPEG Big 80%	N/A*	N/A*	1 every 3 seconds
PNG (16 bit)	2 per second	1 every 1.5 seconds	1 every 4 seconds
SLRAW (16 bit)	5 per second	2 per second	1 every 1.5 seconds
TIFF (16 bit)	5 per second	2 per second	N/A***

* Not tested at this time, sub-optimal performance expected.

** Not supported on this platform.

*** Pending appropriate camera.

**Table 9: 17xx-OEM MicroSD Card Snapshot Performance**

File Format	480p (640x480)	720p (1280x720)	1080p (1920x1080)
JPEG 80%	30 per second	15 per second	7 per second
PNG (color)	9 per second	4 per second	2 per second
PNG (16-bit)	9 per second	3 per second	2 per second
SLRAW (16-bit)	30 per second	24 per second	20 per second
TIFF (16-bit)	30 per second	24 per second	21 per second

Table 10: 17xx-OEM USB SSD Card Snapshot Performance

File Format	480p (640x480)	720p (1280x720)	1080p (1920x1080)
JPEG 80%	30 per second	15 per second	7 per second
PNG (color)	9 per second	4 per second	3 per second
PNG (16-bit)	9 per second	3 per second	2 per second
SLRAW (16-bit)	30 per second	30 per second	29 per second
TIFF (16-bit)	30 per second	30 per second	29 per second

Table 11: 4000-OEM MicroSD Card Snapshot Performance

File Format	SD 640x480	720p (1280x720)	1080p (1920x1080)	4K (3840x2160)
JPEG 80%	30 per second	17.5 per second	10 per second	6 per second
PNG	9 per second	4 per second	2 per second	1 per second
PNG (16-bit)	9 per second	3 per second	2 per second	1 per second
SLRAW (16-bit)	30+ per second	25 per second	14 per second	9 per second
TIFF (16-bit)	30 per second	25 per second	14 per second	9 per second

Table 12: 4000-OEM USB SSD Snapshot Performance

File Format	SD 640x480	720p (1280x720)	1080p (1920x1080)	4K (3840x2160)
JPEG 80%	30 per second	17.5 per second	10 per second	6 per second
PNG	9 per second	4 per second	2 per second	1 per second
PNG (16-bit)	9 per second	3 per second	2 per second	1 per second
SLRAW (16-bit)	30+ per second	30+ per second	25 per second	17 per second
TIFF (16-bit)	30 per second	30 per second	25 per second	17 per second

Table 13: 41xx-OEM MicroSD Card Snapshot Performance

File Format	SD 640x480	720p (1280x720)	1080p (1920x1080)	4K (3840x2160)
JPEG 80%	30 per second	23 per second	10 per second	7 per second
PNG (color)	15 per second	6 per second	3 per second	2 per second
PNG (16-bit)	14 per second	5 per second	3 per second	2 per second
SLRAW (16-bit)	30 per second	29 per second	14 per second	9 per second
TIFF (16-bit)	30 per second	29 per second	14 per second	9 per second

Table 14: 41xx-OEM USB SSD Snapshot Performance

File Format	SD 640x480	720p (1280x720)	1080p (1920x1080)	4K (3840x2160)
JPEG 80%	30 per second	24 per second	11 per second	7 per second
PNG	15 per second	6 per second	3 per second	2 per second
PNG (16-bit)	15 per second	6 per second	3 per second	2 per second
SLRAW (16-bit)	30 per second	30 per second	29 per second	25 per second
TIFF (16-bit)	30 per second	30 per second	28 per second	25 per second



12 Troubleshooting

Issue	Recommendation
System does not appear to save JPEG files.	Make sure that 14-bit digital source images are not being captured. Snapshot of the display image should work.
PNG files appear black in image viewer program.	Most image viewer programs only show the lower 8-bits of a 16-bit PNG. See the EAN-Infrared-Temperature document.
Snapshot features do not appear to be working.	Make sure the Recording software feature has been enabled.
JPEG or PNG files are not on the microSD card.	Check the following: ✓ Card is read/writeable. ✓ Card has been installed correctly. ✓ Card is formatted correctly. ✓ Files are not being saved to an FTP Server.
Issue	Recommendation
JPEG or PNG files are not on the FTP server.	✓ Make sure files are not being save to the microSD card. ✓ Verify the IP address of the FTP server. ✓ Verify the Username and Password. ✓ Make sure the SetSnapshot (0x5E) command was sent to configure the OEM hardware.
Can snapshots be recorded at the same time as recording a video?	Yes, but there may be some performance loss or skipped frames.
Can I record snapshots of the source (capture) image at the same time as the final processed display image?	No. The system can only record either the original capture image or the final display image, but not both at the same time.
Can KLV metadata be recorded to the JPEG/PNG files without the KLV metadata encoding software function?	Yes. Metadata to the OEM hardware can still be sent. The fields will be extracted and added to the EXIF header of the snapshot.
What types of microSD cards can be used?	Sightline recommends the following cards: Kingston SDC10/32GBCP • Kingston 32GB Micro SD / Class 10 • A complete list of tested microSD cards is available in Appendix B.
Should the microSD card be formatted?	Yes. Sightline recommends all microSD cards be formatted before being used in the OEM hardware. See the EAN-Firmware-Upgrade-Utility for more information.
How to check if a microSD card is available using Sightline Protocol.	Use the CurrentSDCardRecordingStatus (0x58) message to check the status of the micros SD card.
The h264 file recorded when using RTP H.264 cannot be played back by the media player or does not play back at the correct speed.	FFmpeg can be used to re-encode the video to include the proper container so playback speed will be correct and compatible with more viewers. For example: <pre>ffmpeg.exe -r 30 -i .\input.h264 -r 30 -f mpegts -vcodec h264 -bf 0 -qp 24 -g 10 output.ts</pre> • The -r argument on the input and output tells FFmpeg what the video rate really is and what is preferred. • The -bf 0 tells FFmpeg to not use b frames. • The qp 24 is the quality of the output, lower is better quality. • The g 10 is the GOP interval.



(Troubleshooting continued)

The *hevc* file recorded when using RTP H.265 cannot be played back by the media player or does not play back at the correct speed.

FFmpeg can be used to re-encode the video to include the proper container so playback speed will be correct and compatible with more viewers. For example:

```
ffmpeg.exe -r 30 -i .\input.hevc -r 30 -f mpegts -vcodec hevc -bf 0 -crf 24 -g 10 output.ts
```

- The *-r* argument on the input and output tells FFmpeg what the video rate really is and what is preferred.
- The *-bf 0* tells FFmpeg to not use b frames.
- The *-crf 24* is the quality of the output, lower is better quality.
- The *-g 10* is the GOP interval.

On the 17xx-OEM, 4000-OEM, and 41xx-OEM the reported frame rate in VLC is different than what the camera is configured for.

Reported frame rate in VLC will be different when using output frame sizes 1920 x 1088 or less. Playback speed will be correct. In Panel Plus use the *Compress* tab to configure a custom *Output Frame Size* to 1920 x 1096 to report accurate frame rate.

12.1 Questions and Additional Support

For questions and additional support, please contact [Sightline Support](#). Additional support documentation and Engineering Application Notes (EANs) can be found on the [Product Support Documentation](#) page in the [Sightline Knowledge Base](#).

Appendix A - Formatting MicroSD Cards and USB Devices

IMPORTANT: All microSD cards and USB devices should be formatted before being used with OEM hardware regardless of card size.

The microSD cards and USB devices can be formatted using MiniTool Partition Wizard available on the MiniTool [website](#). Before starting this procedure, download and install the MiniTool Partition Wizard software.

A microSD card reader is needed for this procedure. If the PC does not have a built-in reader, use an external USB microSD card reader or similar.

A1 MicroSD Card Types

A complete list of microSD cards of varied sizes is available in [Appendix B](#). These have been verified to work with the Sightline OEM processors. Sightline recommends purchasing the exact cards in this table to ensure compatibility and timeliness.

Sightline recommends Class 10 microSD cards. Class 10 cards are rated for a minimum sequential write speed of 10 Mbps or greater.

A2 USB Drive Types

A complete list of USB drives of varied sizes is available in [Appendix C](#). These have been verified to work with the 4000-OEM and 41xx-OEM. Since performance can vary dramatically from device to device, Sightline recommends choosing one of the drives from this table.



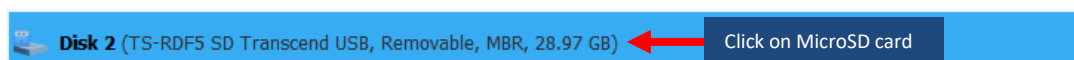
A3 exFAT - 17xx-OEM, 4000-OEM, 41xx-OEM Only

 The following procedure discusses how to format a microSD card as exFAT. The same format can also be used to format a USB drive.

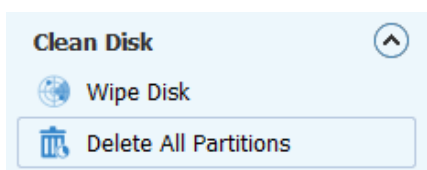
 17xx-OEM recovery requires [FAT32](#).

exFAT (Extensible File Allocation Table) is a filesystem without the limitations of FAT32, which has a max file size of 4 GB. It is also a light-weight filesystem compared to similar systems such as NTFS. Support for the 17xx-OEM, 4000-OEM, and 41xx-OEM was added since most of the USB3 SSD drives that Sightline has evaluated are formatted for exFAT. Using exFAT over FAT32 is not required since we do not recommend recording files over 4 GB, however it is available as an option.

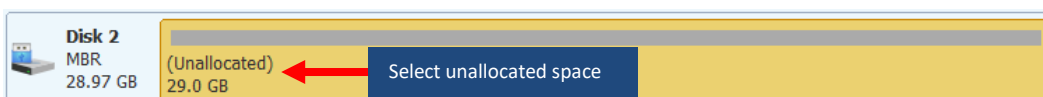
1. Insert the microSD Card into the card reader on the PC.
2. Open the MiniTool application. From the list of drives, highlight the microSD card by clicking on it.



3. From the *Clean Disk* list, click *Delete All Partitions*.



4. From the list of volumes, highlight *(Unallocated)* on the microSD card by clicking on it.



5. From the *Partition Management* list, click *Create Partition*.



6. In the *Create New Partition* window, set the following fields shown in **Figure A3**:

- Optional: *Partition Label* (leave blank for none)
- Create As: *Primary*
- File System: *exFAT*

 Leave partition size at the default value of the card.

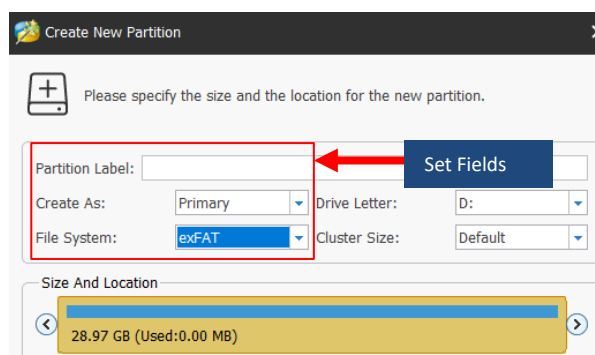


Figure A1: New Partition Window - ExFAT



Setting the cluster size to 1024KB can increase the snapshot rates of 16-bit snapshots (see [Snapshot Rates](#)).

- Click *Apply* from the top menu bar. The operations status screen shows the progress.



- When complete, Click *OK* and close the application. The card is now ready for recording.

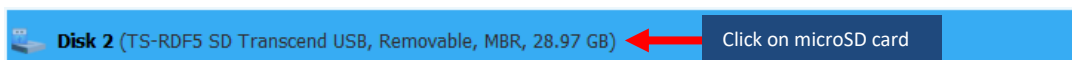
A4 FAT32

The following procedure discusses how to format a microSD card as FAT32. The same format can also be used to format a USB drive.

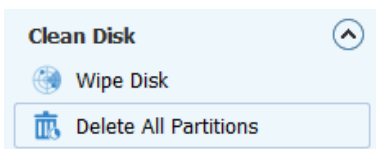
FAT32 (File Allocation Table) is the most used filesystem for removable flash media. FAT32 provides excellent interoperability with other devices and is supported by all modern operating systems.

IMPORTANT: FAT32 is susceptible to power loss induced file corruption during write operations. Consider using smaller file sizes or exFAT if the board is expected to lose power during video recording.

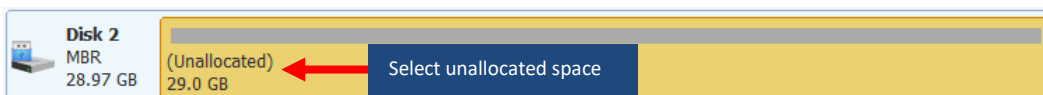
- Insert the microSD Card into the card reader on the PC.
- Open the MiniTool application.
- From the list of drives, click on the microSD card.



- From the *Clean Disk* list, click *Delete All Partitions*.



- From the list of volumes, highlight *(Unallocated)* on the microSD card by clicking on it.



- From the *Partition Management* list, click *Create Partition*.

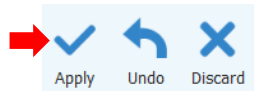




7. In the *Create New Partition* window, set the following fields as shown in **Figure A1**:

- *Partition Label*: Leave blank for none (optional)
- File System: FAT32
- Create As: Primary
- Partition Size: Default value

8. Click *Apply* from the top menu bar. The operations status screen shows the progress.



9. When complete, Click *OK* and close the application. The card is now ready for recording.

Figure A2: New Partition Window - FAT32

Appendix B - Tested MicroSD Cards

Not all Class 10 microSD cards are the same. Different cards have varied read/write performance. Sightline has provided scripts that can help benchmark the microSD card if recording performance is poor. Sightline has evaluated the following microSD cards with the Sightline OEM processors shown in **Table B1**.

Table B1: Tested MicroSD Cards

Manufacturer	Model	Capacity	Speed Class
Delkin	S432FQYFA-U1000-3	32 GB	10, UHS Class 1 MLC
SanDisk	SDSDQ-2048	2 GB	n/a
SanDisk	SDSDQ-4096	4 GB	4
SanDisk	SDSQXCG-032G	32 GB	10, U3, V30, A1
SanDisk	SDSQUAR-064G	64 GB	10, U1, A1
SanDisk	SDSQUAR-128G	128 GB	10, U1, A1
SanDisk	SDSQUAR-400G	400 GB	10, U1, A1
Kingston	SDCIT/8GB	8 GB	10, U1
Kingston	SDC10G2/32GB	32 GB	10, U1
Kingston	SDCS/32GB	32 GB	10, U1
Kingston	SDCG2/32GB	32 GB	10, U3, V30
Samsung	MB-MC256GA	256 GB	10, U3



Appendix C - Tested USB Drives

Different drives have been shown to have different read/write performance. Performance can also vary dramatically with the cable used. When possible, use the cable provided by the manufacturer.

Sightline has evaluated the USB devices shown in **Table C1** with the 17xx-OEM, 4000-OEM, and 41xx-OEM processors. Additional devices will be added once they have been evaluated.

Table C1: Tested USB Devices

Manufacturer	Model	Capacity	Type
Samsung	MU-PA500B/AM	500 GB	SSD
Western Digital	WDBKXH5000ABK	500 GB	HDD

Appendix D - HTTP Live Streaming (HLS)

HLS works by recording short video segments then retransmitting them over HTTP.

IMPORTANT:

- Viewing video with HLS introduces a large amount of latency (~30 seconds or more). The exact amount depends on the current network and system settings.
- HTTP Live Streaming (HLS) requires software version 2.24.xx and higher.

1. Start HLS recording:

- Follow the steps for recording in [Configure Snapshot Source, Destination, and Formats](#) section. Use a filename with extension `.m3u8`.
- Select how often to start a new file. The example shown in **Figure D1** is set at 600 frames (~20 seconds standard video input).
- Click the *Start* button.

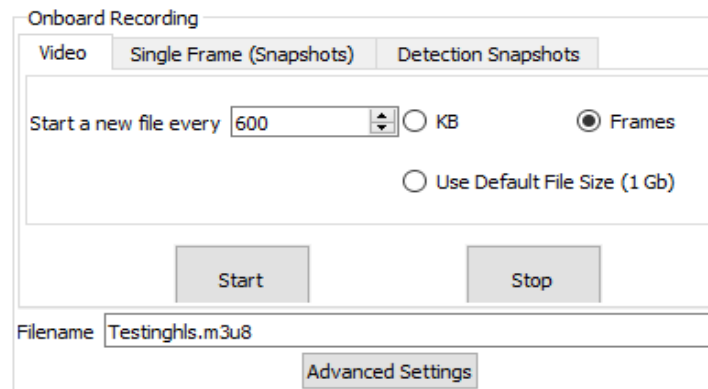


Figure D1: HTTP Live Streaming Example

2. View the stream in VLC (or similar):

- Open a network stream - `http://<<SLA_IP>>/streams/filename.m3u8`.
- The SLA_IP is the IP address of the Sightline hardware. The filename is replaced with the filename specified in step 1.

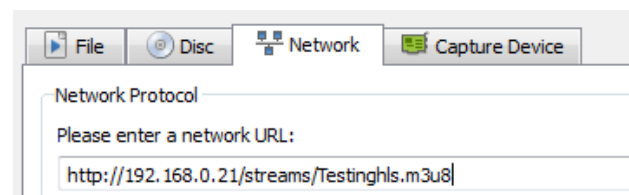


Figure D2: Open Network Stream



Appendix E - EXIF Headers

Table E1: EXIF Headers

EXIF Tag ID	Description	Example / Units	MISB Name	MISB TAG
0x010E	ImageDescription		NA	
0x010F	Make	Sightline Intelligence	Mission ID	3
0x0110	Model	SLA-3000	Platform Designation	10
0x0131	Software	3.1.2	NA	
0x0000	GPSVersionID	2.3	NA	
0x0001	GPSLatitudeRef	{N,S}	SensorLatitude	13
0x0002	GPSLatitude	degrees, min, sec		
0x0003	GPSLongitudeRef	{E,W}	SensorLongitude	14
0x0004	GPSLongitude	degrees, min, sec		
0x0005	GPSAltitudeRef	{0,1}	SensorAltitude	15
0x0006	GPSAltitude	Meters		
0x0007	GPSTimeStamp	Hour,Min,Sec	UTC Precision Timestamp*	2
0x000C	GPSSpeedRef	K - Knots	PlatformIndicatedAirSpeed	9
0x000D	GPSSpeed	Rational		
0x000E	GPSTrackRef	T – True	PlatformHeadingAngle	5
0x000F	GPSTrack	[0.00,359.99]		
0x0012	GPSMapDatum	WGS-84	Image Coordinate System	12
0x0017	GPSTDestBearingRef	T – True	Calculated using Multiple Tags	
0x0018	GPSTDestBearing	[0.00,359.99]		
XMP				
0xE000	SensorBitDepth			
0xE001	Yaw		SensorRelativeAzimuthAngle + platform heading	18 5
0xE002	Pitch		SensorRelativeElevationAngle + platform pitch	19 6
0xE003	Roll		SensorRelativeRollAngle + Platform roll	20 7
0xE007	AboveGroundAltitude			
0xE008	RigName		Platform Call Sign	59
0xE009	RigCameraIndex			

*For all Sightline OEMs GPS time is derived from the onboard Linux time to generate the HHHHHH:MM:SS.sssss from 1980-06-01 as described below:

```
u64 gpsTime = utcTime - (u64)315964811*1000000; //
hours = (u32) (gpsTime / ((u64)3600*1000000));
gpsTime -= (u64)hours *3600*1000000;
minutes = (u32) (gpsTime / (60*1000000));
gpsTime -= (u64)minutes *60*1000000;
seconds = (u32)gpsTime; // seconds with micro-second accuracy
```

Note number of leap seconds.

Unless set by the user, the Linux internal clock will start up for each OEM as follows:

- 1500-OEM: 2000-01-01
- 3000-OEM: 2013-01-10
- 17xx-OEM: 2011-04-05
- 4000-OEM: 2019-02-14
- 41xx-OEM: 2023-09-30



E1 Modify EXIF Header - Use Case Example

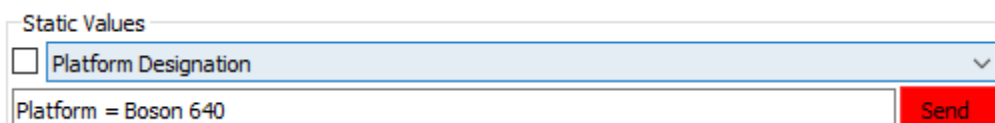
The ability to modify EXIF headers to show the camera make and model can be made through Panel Plus or using Sightline Video Protocol (SVP) commands.

 *The defined string values on the target hardware can be persisted using the LUA script `klvstatic.lua`. Contact [Support](#) for more information to enable this script.*

1. Open Panel plus and establish a connecting to the OEM.
2. From the Panel Plus main menu » *Configure* » *KLV Setup*.
3. In the *Static Values* drop-down menu select *Mission ID* (MISB Tag 3). Enter the camera make in the form of *<Mission ID = Camera make>*. Click *Send*.



4. In the *Static Values* drop-down menu select *Platform Designation* (MISB Tag 10). Enter the camera model name in the form of *<Platform Designation = Camera model>*. Click *Send*.



5. From the *Record* tab, take a [snapshot](#).
6. Save the JPG file to the local PC.
7. Right click on the image and select *Properties*.
8. Click the *Details* tab. Verify that the camera make and model are set as *Mission ID* and *Platform Designation*.

 *SVP commands can also be sent to the target hardware for specific KLV metadata. See the [IDD](#) for more information on the [SLAMetadataStaticValues \(0x14\)](#) command.*

Example -

Elaborate trace of the command for *Camera Make*:

```
MetadataStaticValues{type=0, value={len=24, str="Mission ID = FLIR"}, displayId=0x0002}
```

Example -

Elaborate trace of the command for *Platform Designation*:

```
MetadataStaticValues{type=1, value={len=35, str="Platform Designation = Boson 640"}, displayId=0x0002}
```



Appendix F - Duplicate Camera

This feature requires Dual Processing and Recording software features. For questions and upgrade support contact [Sales](#).

F1 Capture Video Without Overlays Directly to SD Card with One Input

This feature enables users to capture video without overlays directly to the SD card. After downloading the files to the local PC, please forward them to [Sightline Support](#) for post-processing and feedback on parameter optimization.

Before connecting with the Panel Plus software, the OEM board should be powered up and connected.

This example assumes that the customer has configured the camera using Cam 1 and successfully streamed video to Panel Plus.

1. Go to *Configure » Acquisition Settings*. Select a *Camera Index* not in use. In this example *Camera Index* is *Cam 0* where *Cam 1* is the actual camera and *Cam 0* is the duplicate camera.
2. Select *Duplicate* in the *Camera Type* dropdown menu. In the *Duplicate:* dropdown menu select *Cam 1*.

3. Click *Apply* save the parameters and reset the OEM.
4. After the system reboots reconnect to the board. Make sure the board connects and video is streaming from *Cam 1*.

Streaming Camera: 1. Commanded Camera: 1
Frame Rate: 29.88[frames/sec]. Data rate: 2488.96[Kb/sec]

5. From the *Display* tab, select *Cam 0* as the duplicate camera for *Network 1* under *Video Source*. Enable the *Network Display* and *Decode @ P+* radio buttons.

Network Index	Video Source	Network Display		Physical Display		Decode @ P+
		Enable	Resolution	HDMI	Resolution	
0	Cam 1	☒	Out=In	☐	1080p60	☐
1	Cam 0	☒	Out=In	☐	1080p60	☒

6. Verify that video is streaming from the duplicate camera in the status bar. In this example it is from *Cam 0*.

Streaming Camera: 0. Commanded Camera: 0
Frame Rate: 29.98[frames/sec]. Data rate: 2739.38[Kb/sec]



7. If there is no video, go to the Streaming section in the Compress tab and Use My IP - Unicast button for Network 1.

Figure F1: Compress Tab - My IP - Unicast for Network 1

 It is possible to record Cam 0 to the SD card without overlays, while at the same time recording Cam 1 with overlays. This can be done either separately or simultaneously. See the [Video Recording](#) section for more information.

 To stream from the duplicate camera, the actual camera must be in use, either displaying video or sending telemetry.